

Solutions to Bandit Exercises 2.2 and 2.3

Exercise 2.2: Bandit Example

There are $k = 4$ actions. The algorithm uses ε -greedy action selection, sample-average action-value estimates, and initial values

$$Q_1(a) = 0, \quad a \in \{1, 2, 3, 4\}.$$

The observed sequence is

$$A_1 = 1, R_1 = -1, \quad A_2 = 2, R_2 = 1, \quad A_3 = 2, R_3 = -2, \quad A_4 = 2, R_4 = 2, \quad A_5 = 3, R_5 = 0.$$

The question is: on which time steps did the ε case definitely occur, and on which time steps could it possibly have occurred?

Step-by-step action-value estimates

Before each action, compare the selected action with the greedy action under the current estimates.

Time t	Estimates before selecting A_t	Greedy action(s)	Observed A_t
1	$Q_1(1) = Q_1(2) = Q_1(3) = Q_1(4) = 0$	1, 2, 3, 4	1
2	$Q_2(1) = -1, Q_2(2) = Q_2(3) = Q_2(4) = 0$	2, 3, 4	2
3	$Q_3(1) = -1, Q_3(2) = 1, Q_3(3) = Q_3(4) = 0$	2	2
4	$Q_4(1) = -1, Q_4(2) = \frac{1 + (-2)}{2} = -\frac{1}{2}, Q_4(3) = Q_4(4) = 0$	3, 4	2
5	$Q_5(1) = -1, Q_5(2) = \frac{1 + (-2) + 2}{3} = \frac{1}{3}, Q_5(3) = Q_5(4) = 0$	2	3

At time steps 4 and 5, the selected action was not greedy. Therefore the ε case must have occurred at those steps:

$$\boxed{t = 4, 5.}$$

At time steps 1, 2, and 3, the selected action was greedy. Therefore the ε case might or might not have occurred. Even when the ε case occurs, the random selection can still choose a greedy action by chance. Thus the time steps on which the ε case could possibly have occurred are

$$\boxed{t = 1, 2, 3, 4, 5.}$$

If the question asks for the non-definite possible steps, they are

$$t = 1, 2, 3.$$

Exercise 2.3: Long-Run Comparison in Figure 2.2

Figure 2.2 compares three methods on the 10-armed testbed:

$$\varepsilon = 0, \quad \varepsilon = 0.01, \quad \varepsilon = 0.1.$$

In the first 1000 steps, $\varepsilon = 0.1$ performs best in the figure because it explores more aggressively and finds good actions faster. However, the exercise asks about the *long run*. In the long run, the sample-average estimates become accurate, so the main cost is unnecessary exploration.

For a k -armed bandit, once the best action is known, an ε -greedy method selects the optimal action with probability

$$P(\text{optimal action}) = (1 - \varepsilon) + \varepsilon \frac{1}{k} = 1 - \varepsilon \frac{k-1}{k}.$$

Here $k = 10$, so

$$P(\text{optimal action}) = 1 - 0.9\varepsilon.$$

For $\varepsilon = 0.1$,

$$P(\text{optimal action}) = 1 - 0.9(0.1) = 0.91.$$

For $\varepsilon = 0.01$,

$$P(\text{optimal action}) = 1 - 0.9(0.01) = 0.991.$$

Therefore $\varepsilon = 0.01$ is better in the long run in terms of probability of selecting the optimal action. Its asymptotic advantage over $\varepsilon = 0.1$ is

$$0.991 - 0.91 = 0.081,$$

that is,

$$8.1 \text{ percentage points.}$$

Long-run average reward

Let q_* be the true value of the optimal action, and let

$$\bar{q} = \frac{1}{k} \sum_{a=1}^k q_*(a)$$

be the average true action value across all actions. Once the action-value estimates are accurate, the long-run expected reward per step for an ε -greedy method is approximately

$$(1 - \varepsilon)q_* + \varepsilon\bar{q}.$$

Thus the long-run per-step reward advantage of $\varepsilon = 0.01$ over $\varepsilon = 0.1$ is

$$\begin{aligned} & [(1 - 0.01)q_* + 0.01\bar{q}] - [(1 - 0.1)q_* + 0.1\bar{q}] \\ &= 0.09(q_* - \bar{q}). \end{aligned}$$

In the standard 10-armed testbed, the true action values are sampled from $\mathcal{N}(0, 1)$, so $\bar{q} \approx 0$. The expected maximum of 10 standard normal action values is approximately

$$q_* \approx 1.54.$$

Therefore the long-run reward advantage per step is approximately

$$0.09(1.54 - 0) \approx 0.139.$$

So $\varepsilon = 0.01$ eventually obtains about

$$\boxed{0.14}$$

more reward per step than $\varepsilon = 0.1$. Over a long horizon of T steps, this corresponds to a cumulative reward advantage of approximately

$$\boxed{0.14T}.$$

Final answer

In the long run, the best method is

$$\boxed{\varepsilon = 0.01}.$$

It selects the optimal action with asymptotic probability

$$\boxed{99.1\%,}$$

whereas $\varepsilon = 0.1$ selects it with asymptotic probability

$$\boxed{91\%}.$$

Thus $\varepsilon = 0.01$ is better by

$$\boxed{8.1 \text{ percentage points}}$$

in optimal-action probability, and by about

$$\boxed{0.14 \text{ reward per step}}$$

in long-run average reward on the standard 10-armed testbed.